

Hence, acceleration of the object is direction θ and magnitude of acceleration = $g \tan \theta$

- 6 B** Total momentum of wagon remain unchanged because the vertical force by rain does not affect the horizontal momentum of the wagon.
The horizontal speed of wagon decreases because the amount of water in the wagon increases which increases the total mass of the system and so horizontal speed will decrease.

The kinetic energy of the system decreases using the equation $ke = \frac{p^2}{2m}$ since m increases while p remain constant.

- 7 C** The 2 forces, the 2 tensions in the string and the weight of the red need to intersect at a point for